

Name: Student ID:

ECS401: Unit 5 Lab Progress Test

These exercises are to help you judge your progress. It is really important that you make sure you fully understand everything we have covered so far.

Write a fragment of code to do each of the following:

1. Declare an array called `teams` that holds 10 Strings.

Answer:

2. Declare an array called `wins` that holds 5 integers.

Answer:

3. Store the String "Liverpool" in the very first location of array called `teams` as declared above.

Answer:

4. Store the String "Real Madrid" in the last element of array called `teams` as declared above.

Answer:

5. Store the value 2 in all five elements of array called `wins` (using a for loop) as declared above.

Answer:

6. Prints each element of the array `teams` to the screen as declared above.

Answer:

7. Complete the dry run table and work out what is printed to the screen when the following code is executed.

```
int [] a = {1,1,3,1,3,1};          // L1

for(int i = 1; i<=3; i=i+1)        // L2
{
    if (a[i] == 1)                  // L3
    {
        System.out.println(i);    // L4
    }
}
```

Line	a						i	i <= 3 ?	a[i] == 1 ?	print
	[0]	[1]	[2]	[3]	[4]	[5]				

Answer:

Terminology

8. **Arguments** can be passed to a method – they are the pieces of data that the method needs to be given to do its job and are also known as **actual parameters**. Which three different values are passed as arguments to the method `printValue` below.

i) ii) iii)

```
public static void main (String[] param)
{
    int a = 5;
    printValue(a);
    printValue(10);
    printValue(a+2);
    return;
}

public static void printValue (int b) // Print the integer passed
{
    System.out.println("The value is " + b);
    return;
}
```

9. The values passed each time the method is called need to be stored in new variables in the method that uses them. These new variables are called the **formal parameters**. What are the formal parameters of `print2values` below?

i) ii)

iii) Highlight the places where the formal parameters of method `print2values` are declared in the code below.

```
public static void main (String[] param)
{
    int a = 5; int c = 3;
    print2values(2, 4);
    print2values(c, a);
    print2values(a, c);
    return;
}

public static void print2values (int x, int y)
{
    System.out.println("The values are " + x + " and " + y);
    return;
}
```

10. The **return type** of a method is the type of value it passes back (or void if it is a procedure). Highlight the place where the return type of the method add3 below is declared.

```
public static int add3 (int x, int y, int z)
{
    return x + y + z;
}
```